Sulthan MD

IoT and Data — Forward Deployment Engineer

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↗ sulthan2510-iot-de-portfolio.vercel.app

PROFILE

Senior Data Engineer with 6+ years building large-scale data platforms on Azure, now designing and operating Nexus — an IoT energy monitoring platform running in production. I work across the whole stack: commissioning power analysers and industrial routers in the field, building the private network and hardened cloud infrastructure behind them, and modelling the resulting telemetry into analytics that customers act on. Currently completing an M.Sc. in Renewable Energy and Data Engineering at Hochschule Offenburg.

EXPERIENCE

IoT and Data — Forward Deployment Engineer Mar 2026 — present

Nexus · Client: Own platform — live in production

- Designed, built and operate an IoT energy monitoring platform in production, owning the full path from the power analyser in the distribution board to the customer dashboard.
- Commissioned Janitza UMG801 multi-circuit power analysers: CT ratios and connection types per channel, channel-to-asset mapping, Modbus TCP register map, recording intervals and NTP synchronisation across sites.
- Deployed Teltonika RUTC50 (5G) and RUT901 (LTE Cat 4) industrial routers on RutOS with dual-SIM failover, data-limit guards, firewall zoning and fleet-wide remote management and firmware baselines.
- Built a hub-and-spoke WireGuard network giving each building a private encrypted tunnel over CGNAT mobile networks — scoped AllowedIPs, per-device key rotation, no inbound route and no public port at any site.
- Hardened and operate a Hetzner Cloud server: cloud firewall, key-only SSH with fail2ban, a Docker Compose stack (collector, TimescaleDB, Grafana, application, TLS reverse proxy), automated snapshots and tested restores.
- Modelled telemetry in TimescaleDB with hypertables, continuous aggregates, compression and retention policies; built Grafana dashboards for operations and a React dashboard with server-side tenant isolation for customers.
- Developed per-customer energy analytics: predictive maintenance from harmonic distortion, phase imbalance and power-factor drift, plus residual-based anomaly detection against per-circuit baselines by time of day and week.

HiWi Research Assistant Nov 2025 - Feb 2026

Hochschule Offenburg · Client: Research Project - IoT and Data Engineering

- Design and implementation of an End-to-End IoT Pipeline from AHU sensor data to monitoring dashboard for condition and trend analysis.
- Development of a workflow with Node-RED, NATS JetStream, InfluxDB (with Chronograf UI), Grafana and cold Azure Blob Storage (yyyy/mm/dd/hh) including Python analytics for robust, scalable data analysis.
- Operation of the entire pipeline in Docker containers under Linux.
- Setup of GitLab CI/CD pipelines for automated builds, tests and deployments across multiple environments.
- Creation of Grafana dashboards with alerting for AHU monitoring.

Senior Data Engineer

Apr 2022 - Aug 2025

LTIMindtree · Client: Navistar International (USA)

- Developed ETL pipelines in Azure Synapse Analytics to transform and store large-scale data from ADLS Gen2 into actionable insights.
- Data Transformation & Processing
  - Implemented rule engines to perform transformation on raw data from various resources to create processed data.
  - Haversine analysis of vehicles (Destination, Distance travelled) and last communication details of devices and vehicles.
  - Remediation on important features of processed data.
  - File size and count anomaly prediction using Random Forest ML.
  - Data distribution analysis on state and country wise data.
  - Quality controls checks on packets count and packets dictionary.
- Implemented Apache Spark for distributed data extraction, cleaning, and transformation, integrating notebooks into pipelines.
- Improved pipeline efficiency by 50% using Databricks Job Clusters with dynamic scaling and ephemeral provisioning.
- Orchestrated Databricks notebooks from Azure Synapse pipelines to automate and schedule Spark-based workflows.
- Utilized Azure DevOps to implement CI/CD pipelines, ensuring seamless deployment across DEV, UAT, and PROD with version control.
- Created python wheel files to enhance utility compatibility and handled parquet, json and avro formats for efficient data processing.
- Automated pipelines with triggers, configured failure notifications via Azure Logic Apps, and maintained job performance and data integrity.

Data Engineer

Aug 2020 - Apr 2022

LTIMindtree · Client: L&T ECC (India)

- Migrated Spark-Sql-based jobs from Azure HDInsight to Databricks platform, optimizing performance, scalability and cost-efficiency
- Transitioned interactive MS-SQL tables to Azure Delta Tables for improved job efficiency and reliability.
- Designed and configured Azure Databricks pipelines using job clusters.
- Developed and optimized Spark-SQL jobs to efficiently process, transform and analyze daily raw data into meaningful insights.

IoT Engineer

Jun 2018 - Aug 2020

LTI · Client: Tampnet (Norway)

- Research trainee on IoT Sensors, ATEX and IECEx certifications and data retrieval from sensors using NB-IoT and CAT-M networks.
- Engaged in sensors procurement, vendor interaction and lab testing.
- Developed python based APIs to process real-time sensors data into analytical insights for predictive analysis monitoring applications.
- Certifications completion on Python, SQL and Apache-Spark.

EDUCATION

M.Sc. Renewable Energy and Data Engineering

Oct 2025 - Present

Offenburg University of Applied Sciences · Germany

1st Semester | Overall Grade: **1.7**

Bachelor of Engineering in Mechanical Engineering

2014 - 2018

College of Engineering Guindy, Anna University · India

Final Grade: **1.6**

TECHNICAL SKILLS

IoT hardware & field devices
Janitza UMG801 power analysers (Modbus TCP, CT configuration, GridVis, RCM modules), Teltonika RUTC50 / RUT901 industrial routers (RutOS, dual-SIM failover, firewall zones, remote management), Siemens controllers, BACnet, M-Bus, ATEX / IECEx certified sensors

Networking & security
WireGuard VPN (hub-and-spoke, key rotation, CGNAT traversal, MTU tuning), SSH hardening (key-only authentication, disabled root login, fail2ban), firewall zoning, Hetzner Cloud Firewall, TLS and reverse proxy, network segmentation

Cloud infrastructure & Linux
Hetzner Cloud (servers, cloud firewalls, private networks, snapshots and backups), Linux server administration, Docker and Docker Compose, TimescaleDB (hypertables, continuous aggregates, compression, retention), unattended security upgrades

Azure Services - ETL/ELT
Azure Synapse Analytics, Azure Databricks, Azure KeyVault, Azure Logic Apps, ADLS Gen2, Delta Tables, Hadoop, Azure Blob Storage

Big Data & Processing
Apache Spark (PySpark & Spark-SQL), Delta Lake, Kafka, Distributed Computing

Data Monitoring & Visualization
Grafana, InfluxDB, Chronograf

IoT & Real-time Data
MQTT, EMQX, Node-RED, NATS JetStream, Azure IoT Hub and EventHub, WebPubSub, Telegraf, Real-time Streaming

Languages
Python, SQL, C#

Workflow & Orchestration
Apache Airflow, Databricks (Job Clusters & Workflows), Synapse Analytics (Pipelines and Schedulers)

Version Control & DevOps & Containerization
Azure DevOps, GitHub, GitLab CI/CD, Docker

Operating Systems
Linux, MacOS, Windows

CERTIFICATIONS & MODULES

Data Engineer
DataCamp · 2026

Data Engineer Associate
DataCamp · 2026

Azure Databricks workload automation module
Microsoft Azure · 2026

Spark Structured Streaming — incremental processing
Microsoft Azure · 2026

Spark Masters — PySpark and Spark SQL
Udemy · 2024

Apache Airflow
Udemy · 2023

Azure Data Fundamentals — DP-900
Microsoft Azure · 2023

Azure Fundamentals — AZ-900
Microsoft Azure · 2023

Docker & Kubernetes
Udemy · 2024

GenAI — LTIMindtree
LTIMindtree / Coursera · 2024

Data Science and Machine Learning
Coursera · 2023

Python Programming
Udemy · 2023

SQL Mastery
Udemy · 2023

Digital skills self-assessment
Europa.eu · 2024

LANGUAGES

Tamil Native **English** Proficient (C1) **German** B2.2 **Arabic** Reading